

T7 re-proof attempt: in-bundle reproof of [BD-DISCRETE] closes the poly-separated subclass; the residual is [NEST-DEPTH]

TECT verification-first repository · claim B5-BEYOND-LAYER-BOUND

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1. Purpose and scope

The R-035 verdict left two T7 routes for arbitrary- Q DR-2: (i) adopt [BD-DISCRETE] as a standard black box (conditional theorem), or (ii) reprove the discrete well-separated reduction in-bundle to remove the last conditional residual. This note attempts (ii). **Scope:** whether the reduction of an arbitrary finite $Q \subset S^2$ to the separated base case (dr2-decoupling-closure v1.1: separated Q is T6 PROVED CONDITIONAL on the continuous theorem) can be reproved using only elementary arguments plus [BD-CONTINUOUS]. **Not claimed:** any unconditional arbitrary- Q bound; any flip.

2. Pre-registered gates

G2 (success): a complete in-bundle reproof closing arbitrary- Q DR-2 modulo only [BD-CONTINUOUS] $\Rightarrow$ arbitrary- Q DR-2 T6 PROVED CONDITIONAL on the textbook theorem alone, opening the B5 T7 promotion. G1 (partial): the reproof closes only a subclass, or reduces to a named open piece $\Rightarrow$ honest partial, B5 T7 unconditional not achieved. **Outcome: G1.**

3. The poly-separated subclass closes (in-bundle, modulo [BD-CONTINUOUS])

Proposition 1 (Poly-separated closure). *Let $Q \subset S^2$ (antipodal, N points, $\text{diam } Q \asymp 1$ after the affine normalization R-023) have minimal separation $s \geq N^{-C}$ for a fixed C . Then $E_+(Q) \lesssim_{\varepsilon, C} N^{2+\varepsilon}$, modulo only the continuous ℓ^2 -decoupling theorem.*

Proof. Apply ℓ^2 -decoupling at $p = 4$ and scale $\delta = s$, so each $\delta^{1/2}$ -cap carries at most one point of Q ; $f_Q = \sum_a e_a$ has one frequency per cap. With the Besicovitch-mean bridge (R1 = R-034, $M(|f_Q|^4) = E_+(Q)$) and the single-frequency cap norms,

$$E_+(Q) = M(|f_Q|^4) \lesssim_{\varepsilon} \delta^{-2\varepsilon} N^2 = s^{-2\varepsilon} N^2 \leq N^{2+2C\varepsilon},$$

using $s \geq N^{-C}$. Re-labelling ε gives $N^{2+\varepsilon}$. The only non-elementary input is the continuous theorem; no discrete black box and no separation-restriction beyond $s \geq N^{-C}$ enter. $\square$

This is a strict refinement: the bundled [BD-DISCRETE] is here replaced, for the poly-separated subclass, by [BD-CONTINUOUS] alone (machine: assert 1, the exponent $2 + C\varepsilon'$ stays $2 + O(\varepsilon)$ for every fixed C). The crystallographic-shell case ($s \asymp N^{-1/2}$, poly-separated) is an independent proven-case confirmation: $E_+/N^2 \leq 5.27/4.93/3.93$ on $R = 101/909/4994$ (assert 4, R-026 data).

4. The clustered case reduces to [NEST-DEPTH] (the residual)

For sub-polynomially-separated Q ($s < N^{-C}$ for every C) the factor $s^{-2\varepsilon}$ is not $N^{O(\varepsilon)}$ and Prop. 1 fails. The natural move is recursion: a cluster in a ball of radius ρ rescales (R-023, loss-free) to a unit paraboloid patch, and one recurses on the rescaled sub-configuration. The obstruction is the recursion DEPTH.

Lemma 1 (Nesting-depth obstruction). *There exist N -point configurations whose nested-cluster recursion has depth $\asymp \log(1/s)$, unbounded as $s \rightarrow 0$ at fixed N .*

Proof / witness. The lacunary configuration with one-dimensional positions $\{2^{-j} : j = 0, \dots, J\}$ (carried onto S^2) has min-separation $s = 2^{-J}$ and exactly $J = \log_2(1/s)$ dyadic scales at each of which a single point splits off – a genuine split per scale, depth J (machine: assert 2, depth 10/20/40 at $J = 10/20/40$). Iterating the one-step decoupling inequality at a fixed scale across J nested generations carries $D_0^{2^J}$, exponential in the depth (assert 3, $\exp(55.5)$ at $J = 40$) – the R-035 obstruction, now localized to the nesting recursion. $\square$

Consequence. The in-bundle reproof closes the poly-separated subclass but reduces the general case to controlling this depth. Define

[NEST-DEPTH]: a uniform bound on the nested-cluster recursion

depth (equivalently, $s \geq N^{-O(1)}$ suffices).

Then

$$[\text{BD-DISCRETE}] = [\text{BD-CONTINUOUS}] + [\text{NEST-DEPTH}],$$

and the in-bundle reproof discharges the first summand (textbook) but not the second. [NEST-DEPTH] is the genuine open piece; it is the discrete pigeonholing internal to Bourgain–Demeter’s discrete corollary, which the power-sum lemma (R-035) handles for the N -counting but which carries an irreducible D -loss for sub-polynomial clustering.

5. Verdict and the honest ceiling

Gate **G1**: the T7 re-proof does NOT close arbitrary- Q DR-2 unconditionally. What it achieves: (a) the poly-separated subclass at **T6 PROVED CONDITIONAL on [BD-CONTINUOUS]** (textbook decoupling only – a strictly cleaner hypothesis than [BD-DISCRETE]); (b) a clean split of the residual into the textbook half and the single open piece [NEST-DEPTH]. The unconditional record for the fully general case stays $E_+ \lesssim N^{9/4}$ (R-033, modulo [CIRC-INC]). **B5 T7 unconditional is therefore not achieved by this route**; the honest ceiling for arbitrary- Q DR-2 is T6 PROVED CONDITIONAL, and B5 stays T6-conditional on H-NONLATTICE-REMAINDER-EXCLUDED. The remaining genuine routes to B5 T7 are: prove [NEST-DEPTH] (a hard incidence/clustering problem, the open frontier), OR demonstrate that the non-lattice remainder is non-load-bearing for B5’s own statement (the way it is non-load-bearing for the published C_{full} head, synthesis Sec. 4b) – an operator-level structural decision, not a new theorem.

6. Devil’s-advocate / self-adversarial review (CLAUDE.md 6.3)

- (α) “The poly-separated closure is just the separated corollary restated.” VALID with mitigation: it IS the continuous theorem applied at $\delta = s$, but the contribution is the EXPLICIT poly-separation threshold $s \geq N^{-C}$ delimiting exactly where [BD-DISCRETE] is unnecessary, and the resulting clean split $[\text{BD-DISCRETE}] = [\text{BD-CONTINUOUS}] + [\text{NEST-DEPTH}]$. The value is the precise location of the open piece, not new analysis.

- (β) “Maybe [NEST-DEPTH] closes by a cleverer recursion you did not try (broad-narrow, summable- ε).” UPHELD as open: the note does not claim [NEST-DEPTH] is unprovable – only that the elementary telescoping/fixed-scale routes fail (assert 3) and the standard discharge is the Bourgain–Demeter discrete pigeonholing itself. Whether a self-contained in-bundle proof of [NEST-DEPTH] exists is left OPEN, honestly.
- (γ) “The lacunary witness is 1-dimensional; on the curved S^2 curvature might forbid the deep nesting.” VALID with mitigation: the witness is carried onto a great circle / low-curvature direction where the nesting survives at the relevant scales; curvature helps the SEPARATED case (it is what makes decoupling work) but does not bound the nesting depth of a clustered set, which is a combinatorial scale count. A curvature-exploiting depth bound would be exactly a proof of [NEST-DEPTH].

Quantitative sanity check (mandatory). The poly-separated exponent $2 + C\varepsilon'$ is monotone in C and equals $2 + O(\varepsilon)$ for fixed C (assert 1: 2.2/2.4/2.6/3.0 at $C = 1/2/3/5$); the lattice ratios are bounded and DECREASING ($5.27 \rightarrow 3.93$, assert 4), the signature of $N^{2+o(1)}$ on a proven poly-separated family; the nesting depth equals $\log_2(1/s)$ exactly (assert 2) and its fixed-scale loss is exponential (assert 3). All five checks PASS, exit 0.

7. Result footer

Result ID:	B5-BEYOND-LAYER-BOUND / DR-2 T7 re-proof attempt ([BD-DISCRETE] in-bundle)
Precise statement:	T7 RE-PROOF ATTEMPT, gate G1 (HONEST PARTIAL). Poly-separated subclass ($s \geq N^{-C}$): $E_+(Q) \leq \{\varepsilon, C\} N^{\{2+\varepsilon\}}$, PROVED in-bundle modulo only the continuous l2-decoupling theorem [BD-CONTINUOUS] (one frequency per s-cap + Besicovitch bridge R-034). General case: reduces to [NEST-DEPTH] (nested-cluster recursion depth; unbounded for lacunary clusters, depth = $\log_2(1/s)$). Split: [BD-DISCRETE] = [BD-CONTINUOUS] + [NEST-DEPTH]; the reproof discharges the textbook half only.
Scope:	arbitrary-Q DR-2 reduction. Poly-separated closes; clustered open. Unconditional general record stays $N^{\{9/4\}}$ (R-033).
Evidence grade:	PARTIAL ADVANCE (subclass closed modulo [BD-CONTINUOUS]; residual [NEST-DEPTH] named). dr2_t030_bd_discrete_reproof.py 5/5.
Reproduction command:	python codes/vacuum/dr2_t030_bd_discrete_reproof.py
Falsifier:	a poly-separated Q with $E_+ > N^{\{2+\varepsilon\}}$; a bounded-depth proof for lacunary clusters (would CLOSE [NEST-DEPTH] $\rightarrow$ upgrade); a lattice family with E_+/N^2 unbounded.
Tier before / after:	none -- operator decides. B5 stays T6-conditional; arbitrary-Q DR-2 stays T6 PROVED CONDITIONAL; T-030 unconditional OPEN; B5 T7 NOT achieved by this route.
No-overclaim statement:	the unconditional reproof does NOT close; only the poly-separated subclass closes (modulo the textbook theorem); [NEST-DEPTH] is open; no flip.
Next required action:	operator: (a) adopt [BD-CONTINUOUS] as the standard import and register arbitrary-Q DR-2 as T6 PROVED CONDITIONAL with the residual

narrowed to [NEST-DEPTH]; or (b) pursue
[NEST-DEPTH] as a frontier problem; or (c)
take the non-load-bearing structural route to
B5 T7 (synthesis Sec. 4b).